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The effect of multiple polarising lenses on
intensity of light and Brewster's angle
measurement

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1 Introduction

Polarisation is a phenomenon widely documented throughout many fields, such as the field of physics and optics. The application of polarisation is most commonly seen in glasses, as glasses with polarising lenses reduce glare formed from flat surfaces, such as flat bodies of water. This improves the overall comfort and quality of life of the wearer [2].

While the application of polarising lenses seems familiar already, what this experiment aims to examine is the effects and relationships that multiple polarising lenses have upon a beam of light. This practical is split into 3 sections that focus on different aspects of polarising lenses. The first section being that of a beam of light passing through 2 lenses, then in the next a beam of light passes through 3 polarising lenses. The final section closely examines the Brewster's angle.

Brewster's angle describes the angle at which any depolarised light is incident on a surface, such that part of the reflected light is reflected as polarised light[4]. This law is also used in the development of glasses with polarising lenses. The theory surrounding these phenomena shall be covered with more depth in the theory section of this report.

The aim of this experiment is unique for each subsection. For the first section, the aim is to measure the intensity of the light that passes through the 2ND polarising lens. For the 2ND section, the aim is again to measure the intensity, but through the 3rd lens instead of the 2ND lens. For the final section, the aim is to measure the Brewster's angle of the material used within this experiment.

2 Theory

This theory section shall be divided into 4 sections. The first being a general coverage of necessary background theory, and then the 3 other sections delving deeper into that section's theory respectively.

2.1 General theory of polarising light

Light is a type of electromagnetic wave. These waves have two components to them: the electric field component and the magnetic field component. Mathematically, both these components can be expressed as:

$$E = E_m \sin(kx - \omega t) \quad (1)$$

This being the electric field wave, and

$$B = B_m \sin(kx - \omega t) \quad (2)$$

being the magnetic field wave. E_m and B_m both refer to the maximum amplitude that the respective waves can reach. Note that as the electromagnetic wave travels, the wave propagates such that the direction of propagation is perpendicular to both the electric component and the magnetic component.[7].

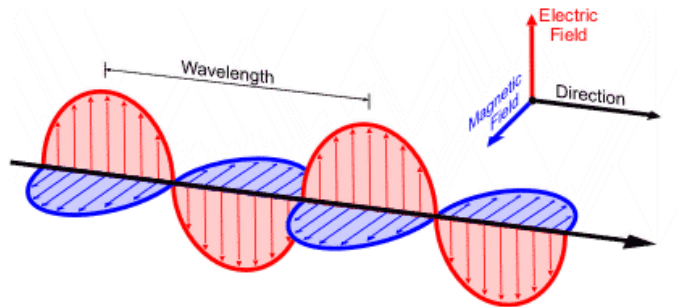


Image source: <https://www.noaa.gov/jetstream/satellites/electromagnetic-waves>

The intensity of the electromagnetic wave can be described as the rate of change of energy in a given area over a given time. The derivation of the equation that describes the intensity of an electromagnetic wave utilises equation (1) mostly, as our modern technology measures electric fields easier than magnetic fields [7]. The equation describing intensity is

$$S = \frac{1}{c\mu_0} E^2 \quad (3)$$

However, if the intensity of a full cycle of a field is considered, as well as taking into consideration the root means square of the electric field, the equation then derives into:

$$I = \frac{1}{c\mu_0} E_{rms}^2 \quad (4)$$

Polarisation occurs when an electromagnetic field only oscillates in one plane. For example, if a field is subjected to a polarising lens, the magnetic field may be absorbed by the material, leaving only the electric in the plane [7].

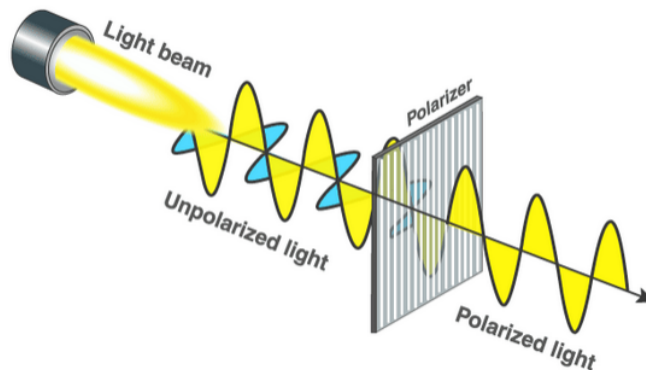


Image source: <https://byjus.com/physics/polarization-of-light/#what-is-polarization>

Typically, sources of light (such as an incandescent bulb or the sun) emit electromagnetic waves with many orientations when compared to one another. This light is then

defined as depolarised light, typically depicted as scatter or "chaotic" light waves.

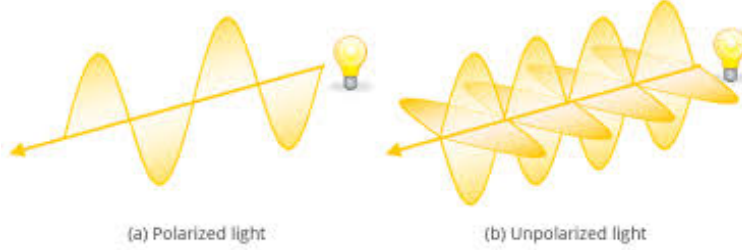


Image source: <http://www.bigshotcamera.com/learn/lcd-display/polarization>

Depolarised light can be transformed into polarised light using an object called a "Polarising sheet". or Polaroids more commonly (invented by Edwin Land in his years as an undergraduate student [7]).

2.2 Two polarizers

Due to the nature of a Polaroid as well as a polarised wave, the intensity is not the same as a normal electromagnetic wave as described by equation (3). Rather, the intensity can be described with the following equation:

$$I = \frac{1}{2}I_0 \quad (5)$$

This equation is so to say an approximation of the intensity of a polarised wave that was originally an depolarised light before being transformed by the Polaroid. There is an alternative equation, only applicable to polarised light being polarised again by a Polaroid. This equation is:

$$I = I_0 \cos^2(\theta) \quad (6)$$

Where I_0 is the initial intensity of the wave, and θ is the angle of the polarisation direction when compared to the approaching wave. Equation (6) will be used to describe the intensity of the second time's polarised wave.

2.3 Three polarizer

For this experiment, the three polarizers are orientated as follows: the first and the last having polarisation directions orthogonal to one another. The 2nd polarizer is then placed in between and set at a certain angle ϕ . Depolarized light passes through the first polarizer. The intensity of the first polarized light is I_1 . Then, the light passes through the 2nd polarizer and the intensity of the newly polarized light is I_2 , which can be described with equation (6).

However, the angle of the third polarizer when compared to the 2nd polarizer can be described as $\frac{\pi}{2} - \phi$. Thus, the intensity of the light passing through the third polarizer, I_3 , can be described as:

$$I_3 = I_2 \cos^2\left(\frac{\pi}{2} - \phi\right) \quad (7)$$

If we substitute the equation of I_2 into the equation, we then get:

$$I_3 = I_1 \cos^2(\phi) \cos^2\left(\frac{\pi}{2} - \phi\right) \quad (8)$$

The equation, after applying trigonometric identities to the equation, then becomes:

$$I_3 = \frac{I_1}{4} \sin^2(2\phi) \quad (9)$$

Due to the nature of this set up, and because the experiment starts with the intensity I_3 being at a maximum, ϕ can be related to an angle θ such that

$$\phi = 45^\circ + \theta \quad (10)$$

2.4 Brewster's angle

Brewster's law states that "the maximum polarization (vibration in one plane only) of a ray of light may be achieved by letting the ray fall on a surface of a transparent medium in such a way that the refracted ray makes an angle of 90° with the reflected ray"[6].

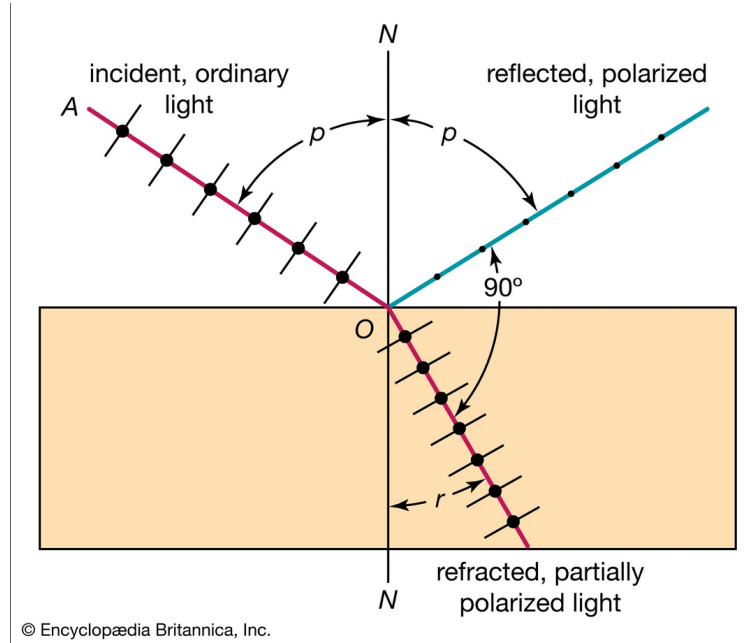


Image source: <https://www.britannica.com/science/Brewsters-law> Mathematically, Brewster's law can be described using the following equation, derived from Snells law:

$$\tan(\theta_p) = \frac{n_2}{n_1} \quad (11)$$

Where n_1 is the index of refraction where the beam of light starts, and n_2 is the index of refraction where the beam of light submerges into.

3 Procedure

Similar to the theory section, the procedure shall be divided into 3 sections. The first section focusing on the two polarizers experiment, the second focusing on the 3 polarizers experiment, and the last focusing on the Brewster's angle.

3.1 Two polarizers

1. Set up a beam of light such that it shines directly onto an intensity sensor.
2. Set up the first polarising lens on a mount such that the beam of light passes through it.
3. Set up a second polarising lens that can have its polarization direction rotated. Place it in front of the preceding polarising lens.
4. To start intensity measurement, adjust the lens of the second polariser such that the angle (with respect to the first polarizer) is at 0° .
5. Begin recording the intensity and slowly rotate the lens of the second polarizer.
6. Ensure that the measuring equipment is functioning correctly and data is measured correctly.

3.2 Three polarizers

1. Position the beam of light such that it will shine onto the detector.
2. Place the first polarising lens in front of the beam.
3. Place the second polarising lens in front of the first one, but leave space for an additional lens.
4. Place the third polarising lens with an adjustable lens such that it can have its polarization direction rotated.
5. Ensure that the first and the last lenses have perpendicular polarization directions.
6. Begin recording data and slowly rotate the lens of the second polarizer.

3.3 Brewster's angle

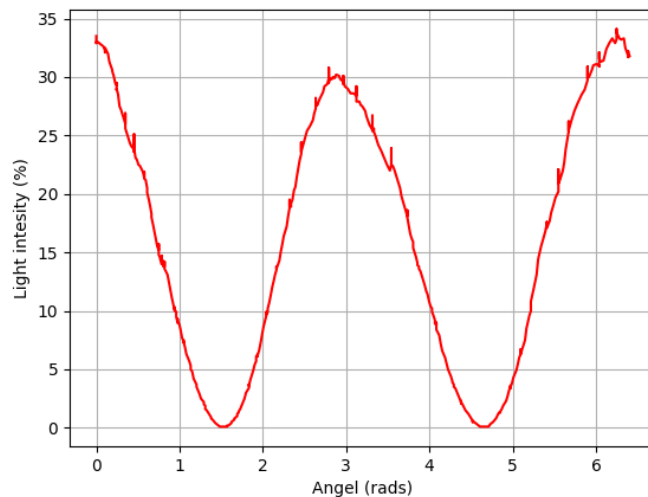
1. Place an acrylic semicircular lens atop a platform that can rotate. Ensure that the platform is at a height such that a beam of light should be able to shine onto the lens and the intensity detector adjacent to the lens.
2. Rig the intensity measuring device such that it can swivel around the acrylic lens. This is to capture polarising rays from a multitude of angles.
3. Start the set-up with the beam and lens being positioned such that the beam barely touches the flat side of the semicircle. Record the intensity.

4. Rotate the lens a few degrees anti-clockwise ($\pm 5^\circ$). Repeat the experiment, but now move the rotating arm of the intensity measuring device until it again captures the reflected ray. Record intensity.
5. Repeat this process until an intensity of 0 is reached.

4 Results

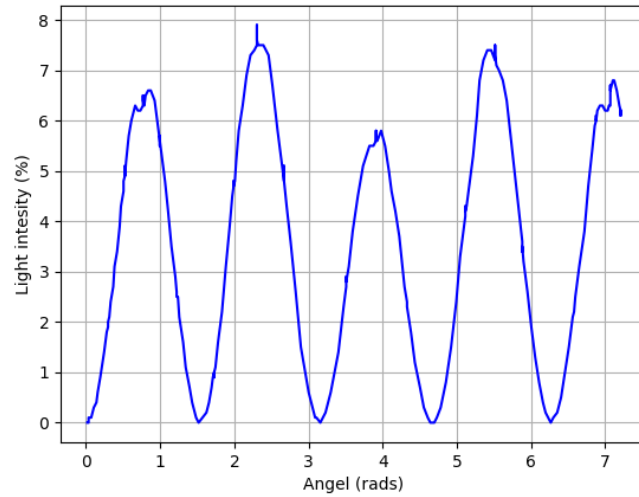
The following section is just the recorded data of the experiments shown onto 3 graphs. The first graph being that of the two polarizer tests, the second being that of the three polarizer test and the last being that of the Brewster's angle test. The data was plotted in Python using the pandas and Matplotlib packages.

4.1 Two Polarizers



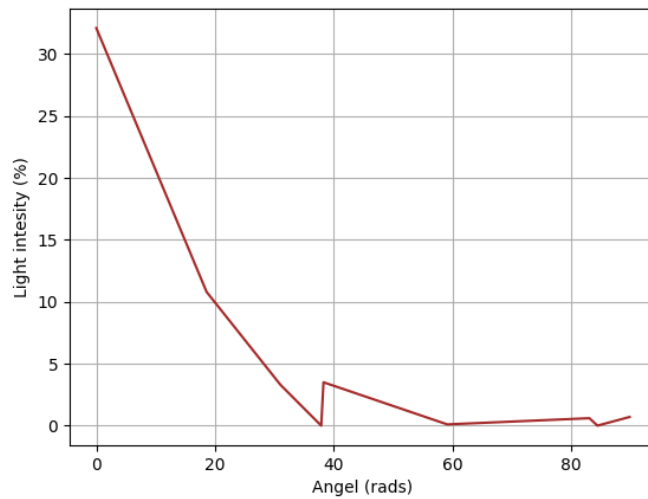
Graph 1: Intensity of light after exposure to two polarising lenses.

4.2 Three Polarizers



Graph 2: Intensity of light after exposure to three polarising lenses.

4.3 Brewster's Angle Measurement



Graph 3: Intensity of the reflected light that shines through a polarising lens.

5 Discussion

This discussion will follow a similar structure to that of the results, where it will be split into smaller discussions analysing the results of each section.

For the two polarizers test, it exhibits a clear cosine graph with fluctuations in intensity, as predicted by formula (6). The initial intensities are repeated at the crests of the cos graph, while continual 0 points are also found at the troughs. When looking at the angle, it is clear to see that the intensity was 0 when the angle was 0 rad. There are odd, small "spikes" in the graph, but this could be attributed to the individual rotating the lens at a non-constant rotational velocity.

For the three polarizers test, the intensity of the peaks is significantly less than that of the two polarizers trial. In the two polarizers test, the intensities peaked at nearly 35% intensity, where here it only peaks at nearly 8%. This could be attributed to the relationship between the angles of the three polarizers. Examining the shape of the graph shows a clear sin graph shape, where it starts at an intensity of 0% instead of at a higher value (like that of the previous experiment). The graph, however, does show more inconsistencies with its peak intensities. While that of the previous experiment was consistently near 35% intensity, this fluctuates more from the 8% intensity.

For the Brewster's angle experiment, one can see that it is a straight line with a negative gradient value, with a then slight pick up in intensity at the point just before 40° . Upon examining the graph closer, one can see that when the intensity is closest to 0, the angle is at $\pm 37.9^\circ$. Although user error may have led to the increased spike read after that, this data of intensity seems to be accurate enough for an accurate data estimation.

6 Conclusion

The aim of this experiment was as follows: to determine intensities of light after passing through second and third polarising lenses, and then finally to determine the Brewster's angle of an acrylic half-sphere.

While the data set does have some minor inconsistencies at times, the overall collected data is well in agreement with the mathematical modelling used to theoretically explain these phenomena. Especially with regard to the first two experiments within this experiment. This shows the reliability of the mathematical models derived for polarising lenses and shows also the effectiveness of this invention in reducing intensity of light. Its application in not just glasses for people to wear, but also in technology such as T.V. screens and other scenarios is filled with potential.

As for the Brewster's angle segment, this experiment greatly shows the effectiveness of calculating the Brewster's angle for a given material. This can then be used again for other similar transparent objects to then calculate not only their Brewster's angle but also their index of refraction. If the item's index of refraction is already known, and the set-up is somehow placed in a material with an unknown index of refraction, then this method can also be used to calculate the index of refraction.

Overall, all these experiments yielded quite accurate results that match the mathematical models well.

7 Additional Questions

Question 1: Name four ways in which light may be polarized.

Light can be polarized in the following ways:

- Through the use of Polaroids (or polarizing materials).
- Using transparent surfaces and utilizing Brewster's law to get the correct angle for polarization.
- Refracting a beam of light to get a partially polarized beam of light.
- Scattering light when it passes through a medium, partially polarizes it as well [3]

Question 2: What is circularly, elliptically and linearly polarized light?

Linearly polarized light is polarized light that propagates in the shape of a plane.

Circularly polarized light is polarized light that consists of two plane waves that are out of phase by 90° . [5]

Elliptically polarized light is polarized light of two planes that have varying amplitudes and that either are out of phase by 90° or another angle.

Question 3: What is Brewster's angle?

As stated in the theory section, "the maximum polarization (vibration in one plane only) of a ray of light may be achieved by letting the ray fall on a surface of a transparent medium in such a way that the refracted ray makes an angle of 90° with the reflected ray" [6].

Question 4: State and describe 4 applications of polarization.

Some applications of polarization are the following:

- It is used in the lenses of glasses to reduce glare and strain on the eyes.
- Polarized screens are used in devices such as tablets to make the screen easier to see in the daylight. [1]
- It is also used in medical equipment, such as Scanning Laser Polarimetry (SLP). This measures the thickness of certain nerve fibres. [8]
- Blue light filters on glasses are also polarizers and are used often for individuals who are near computer screens often. [8]

Question 5: Given m polarizers, each one rotated by x degrees with respect to the previous one's polarization axis, derive the general relation between the number of polarizers, m , and the fraction of transmitted light, I_m/I_0 . The original polarized wave has an intensity I_0 , and the first polarizer, $m = 1$, is rotated x degrees with respect to the polarization axis of this wave.

To start, we first notice that the intensity of light with 2 polarizers is:

$$I_2 = I_1 \cos^2(\theta) \quad (12)$$

Introducing a 3rd polarizer yields:

$$I_3 = I_2 \cos^2(\theta + x) \quad (13)$$

Substituting the first equation into the second yields:

$$I_3 = I_1 \cos^2(\theta) \cos^2(\theta + x) \quad (14)$$

A general series now appears. If we let n be the number of polarizers, and m be the amount of intensities of light being measured, we have that:

$$I_n = I_1 \cos^2(\theta) \cos^2(\theta + x) \dots \cos^2(\theta + (n - 2)x) \quad (15)$$

Thus the ration between I_n to I_1 is :

$$\frac{I_n}{I_1} = \cos^2(\theta) \cos^2(\theta + x) \dots \cos^2(\theta + (n - 2)x) \quad (16)$$

Question 6: Polarized light with intensity I_0 passes through three polarizers. Each polarizer makes an angle with respect to the previous polarizer's polarization axis which is half the angle of that polarizer with respect to its predecessor's polarization axis. (Let the angle of the first polarizer with respect to the polarization axis of the polarized wave be x). Solve for x using a graphical method, given that $I_3/I_0 = 0.5$. Attach a graph indicating your solution.

8 References

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